

BOARD PITCH | KENYA PIPELINE COMPANY (KPC)

Stopping Pump Failures Before They Happen

Predictive maintenance for KPC's pipeline pumps: a KES 5 million, 3-month pilot

Team NULL_TERMINATORS | Week 11

Presented by Eugene Obunde Okumu

We are asking the Board to approve a one-station pilot

Approve **KES 5 million** and **3 months** for a pilot at **one station**. If it works, roll out to all stations for a further KES 10 million.

KES 5M

Pilot: one station, three months.
This is the most the Board puts at risk.

KES 10M

Fleet-wide rollout, released only if the pilot meets its targets.

~7 months

Payback at full scale, on KES 25.1M net savings a year.

The rest of this pitch shows the problem, the solution, the numbers, and how we protect the Board's money.

THE PROBLEM

Pumps are serviced by the calendar, not by their condition

KPC's pumps push fuel through **1,342 km** of pipeline. Servicing follows a fixed schedule:

- Healthy pumps are serviced needlessly.
- Worn pumps can fail between visits.
- A failure halts deliveries, forces costly emergency repairs and can lead to leaks.

KES 63M

2018 Kiboko spill: 551,000 litres of fuel lost, in product alone.

KES 14M

Fuel lost from a hairline leak left unnoticed for a single day.

KES 103M+

2011 Sinai fire: began with a faulty part worth under KES 1,000. About 100 people died.

Low-cost sensors that tell planners which pump needs attention next



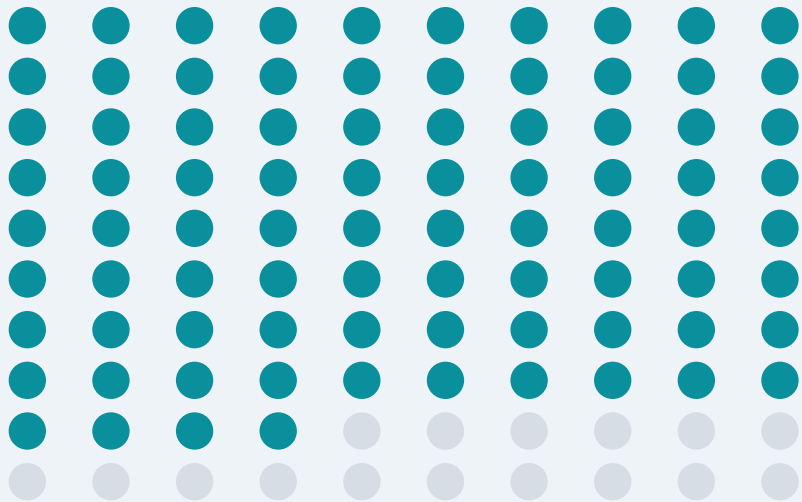
Weather and local activity around each station are factored in.

In a trial on simulated data, it caught 84% of failures

84%

of failures caught early

Picture 100 failures: 84 are flagged in time.

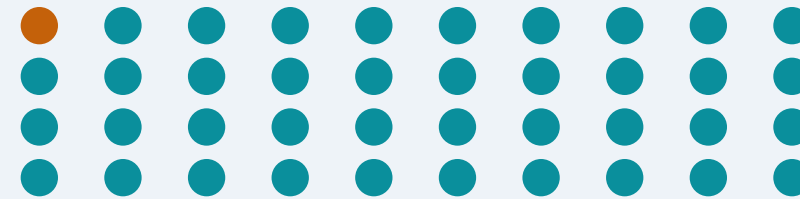


● Caught early

2.4%

false alarms

About 1 in 40 alerts.



● Correct alert

● False alarm

Prototype trial on **simulated data for 24 pumps** (Week 6).
Dots are an illustration of the rates.

Honest caveat: these results are simulated. Confirming them on KPC's real pumps is the purpose of the pilot.

What a planner gets every morning

RISK

Chance of failing in the next 7 days

Each pump is rated daily, so attention goes where it is needed.

TIME LEFT

Estimated hours of life remaining

Repairs can be planned instead of forced by an emergency.

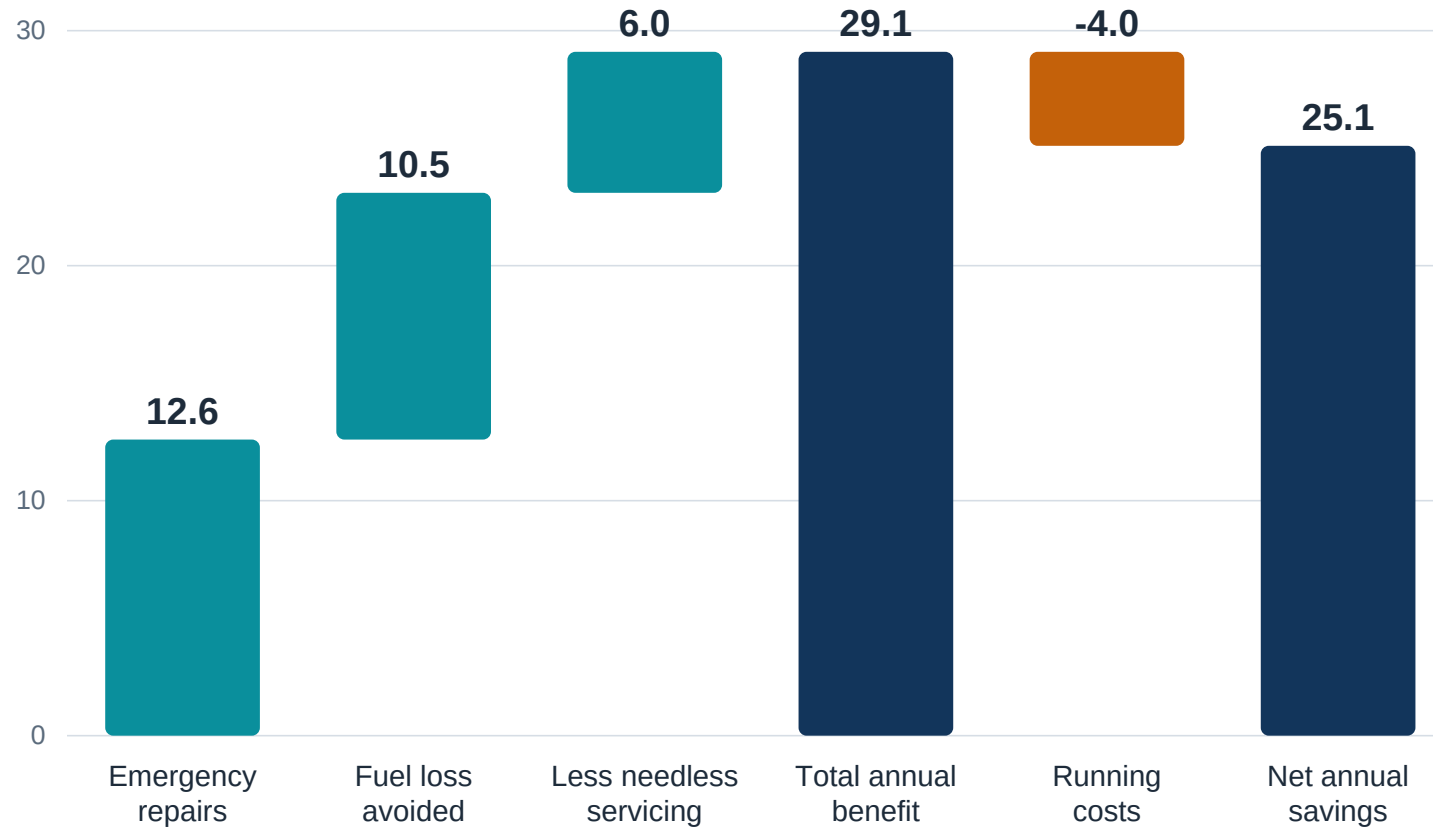
ONE LIST

A single ranked repair list

One screen for planners, with email alerts.

Built for trust: every alert says in plain words why a pump is flagged and how many hours it has left. During the pilot it runs alongside the current schedule.

Net savings of KES 25.1 million a year



The math

Emergency repairs

6 failures/yr x 84% caught early x (KES 4.0M emergency - KES 1.5M planned) = **12.6**

Fuel loss avoided

One Kiboko-size loss (KES 63M) every 3 years = KES 21M/yr x 50% prevented = **10.5**

Needless servicing

60 services/yr x 25% skipped x KES 0.4M each = **6.0**

Running costs

Cloud, sensor upkeep, support = **-4.0**

Fleet size (30 pumps), failure count and repair costs are planning assumptions, replaced with KPC actuals early in the pilot.

KES 15 million in, back in about 7 months

KES 15M

One-off investment

KES 5M pilot + KES 10M rollout.

Of the KES 15M in total: 30 pumps x KES 0.25M sensors = 7.5M, plus setup and training = 7.5M.

~7 months

Payback period

$15.0 / 25.1 = 0.6$ years
Investment divided by net annual savings.

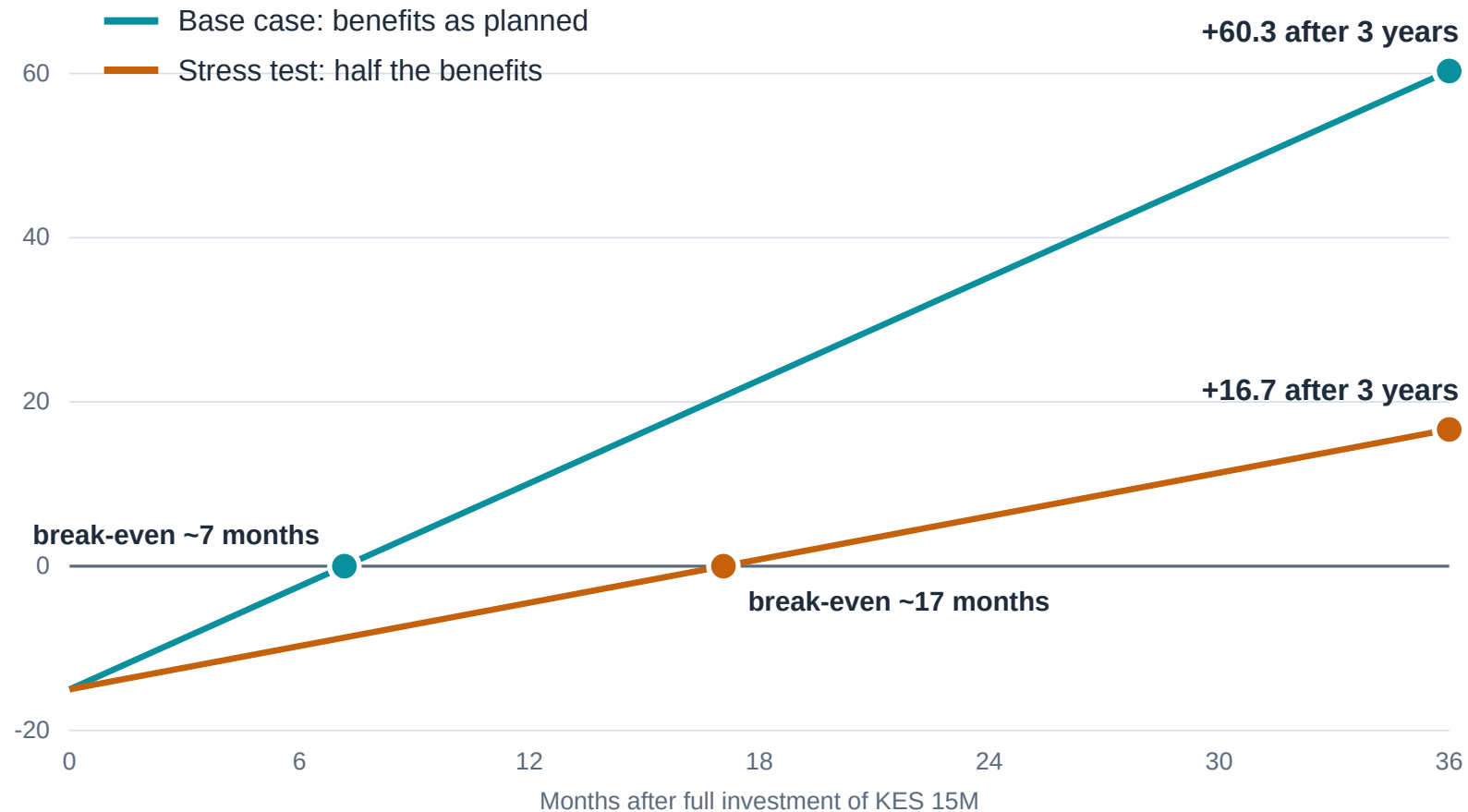
223%

3-year return on investment

$(87.3 - 27.0) / 27.0$
Benefits $3 \times 29.1 = 87.3$.
Costs $15.0 + 3 \times 4.0 = 27.0$.

Fuel value uses KPC's documented Kiboko incident and EPRA's 2026 fuel costs.

Even if benefits are only half, the pilot still pays back



If benefits are half

Payback still about **17 months**.

3-year return still **62%**.

Not counted

Clean-up costs, legal claims and lives protected. Each could exceed the figures above.

Cumulative net = -15.0 + net annual savings x years. Base 25.1/yr; half-benefit case 10.55/yr.

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The two biggest risks, and how we handle them

Risk 1: Not yet proven on KPC's real pumps

Our 84% figure comes from simulated data, so real-world accuracy could be lower.

OUR RESPONSE

Test at one station first, so only KES 5M is at stake. The remaining KES 10M is released only if the pilot catches at least 80% of failures with no more than 5% false alarms.

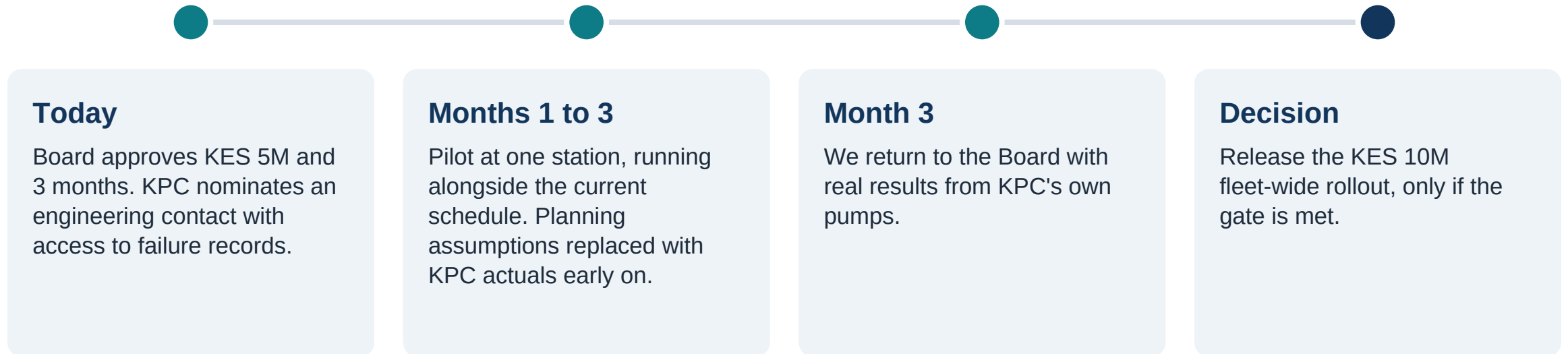
Risk 2: Planners may not trust or act on alerts

Data access may also be slow.

OUR RESPONSE

Every alert says in plain words why a pump is flagged and how many hours it has left. We run alongside the current schedule during the pilot. We agree a KPC engineering contact and data access before starting.

Pilot first, then decide: the KES 10M is gated



The gate: the rollout proceeds only if the pilot catches **at least 80%** of failures with **no more than 5%** false alarms. If it does not, the Board's total exposure is KES 5 million.

RECOMMENDATION

GO: pilot first, then decide

- 1 Approve KES 5 million and 3 months for a one-station pilot.
- 2 Nominate a KPC engineering contact with access to historical pump failure records.
- 3 At the end of the pilot we return with real results. The KES 10 million rollout goes ahead only if the targets are met.

THANK YOU

Questions

Team NULL_TERMINATORS, presented by Eugene Obunde Okumu

Quick reference

Ask: KES 5M, 3 months, one station. Rollout: KES 10M, gated.

Net savings KES 25.1M a year. Payback ~7 months. 3-year return 223%.

Half-benefit stress test: payback ~17 months, 3-year return 62%.

Gate: at least 80% of failures caught, no more than 5% false alarms.

Sources: Breaking Kenya News (2020); Pipeline and Gas Journal (2019); Nation (2020); EPRA (2026); team prototype results (Week 6).